

Institute of Actuaries of India

**Subject CT5 – General Insurance, Life and Health
Contingencies**

April 2016 Examinations

INDICATIVE SOLUTIONS

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) Lives that are selected from a larger group of people, in some non random way with respect to mortality, will experience different mortality at any given age from the group as a whole. Such lives are then referred to as having select mortality. [1]

ii)

$$\begin{aligned}
 {}_{20}p_{[25]:[30]} &= {}_{20}p_{[25]} * {}_{20}p_{[30]} \\
 &= \frac{l_{45}}{l_{[25]}} * \frac{l_{50}}{l_{[30]}} \\
 &= (9801.3123 * 9712.0728) / (9951.9913 * 9923.7497) \\
 &= 0.963852
 \end{aligned}$$

[2]
[3 Marks]

Solution 2:

- i) Main forms of selection are
 Temporary initial selection
 Class selection
 Time selection
 Adverse selection
 Spurious selection [2]

ii)

A change in underwriting level may alter effect of temporary initial selection. For example, rigorous underwriting leads to increased intensity and duration of temporary initial selection.

This may give the impression that mortality rates are improving faster than they really are. Spurious selection is occurring because the effect of time selection is getting distorted.

Conversely, relaxation of underwriting lessens the effect of temporary initial selection. This gives the impression that mortality rates are changing at a slower rate than they really are. This again distorts the effect of time selection. [2]

[4 Marks]

Solution 3:

$${}_{1.4}q_{54.5} = 1 - {}_{1.4}p_{54.5}$$

Now

$${}_{1.4}p_{54.5} = {}_{0.5}p_{54.5} * {}_{0.9}p_{55}$$

$${}_{0.5}p_{54.5} = 1 - {}_{0.5}q_{54.5}$$

Now by UDD:

$${}_{0.5}q_{54.5} = 0.5 q_{54} / (1 - 0.5q_{54})$$

$$\begin{aligned} {}_{0.5}q_{54.5} &= 0.5 * 0.00714 / (1 - 0.5 * 0.00714) \\ &= 0.00358279 \end{aligned}$$

$$\text{Hence } {}_{0.5}p_{54.5} = 0.99641721$$

$$\text{Also, } {}_{0.9}p_{55} = 1 - {}_{0.9}q_{55}$$

$$= 1 - 0.9 q_{55}$$

$$= 1 - 0.9 * 0.00797$$

$$= 0.992827$$

Hence:

$$\begin{aligned} {}_{1.4}q_{54.5} &= 1 - {}_{0.5}p_{54.5} * {}_{0.9}p_{55} \\ &= 0.01073 \end{aligned}$$

[3 Marks]

Solution 4:

Let P be the annual advance premium.

$$\ddot{P}a_{x:n} = 50,00,000 * A_{x:n}^1 + n * P A_{x:n}^1$$

$$\ddot{P}a_{35:30} = 50,00,000 * A_{35:30}^1 + 30 * P A_{35:30}^1$$

Using 6% Interest rate and AM92 Ultimate mortality

$$P * 14.350 = 50,00,000 * (A_{35} - v^{30} * {}_{30}p_{35} * A_{65}) + 30 * P * v^{30} * {}_{30}p_{35}$$

$$\text{Now } A_{35} = 0.09488 \text{ and } A_{65} = 0.40177$$

$$P * 14.350 = 50,00,000 * (0.09488 - 1.06^{(-30)} * (8821.2612/9894.4299) * 0.40177)$$

$$+ 30P * 1.06^{(-30)} * (8821.2612/9894.4299)$$

$$P * 14.350 = 50,00,000 * 0.03251 + 4.657 P$$

$$9.693 P = 1,62,550$$

$$P = \text{INR } 16,770$$

[6 marks]

Solution 5:

- i) Present value of an immediate annuity of 1 per annum payable continuously during the lifetime of life now aged x is denoted by $\bar{a}_{T_x}$ where T_x is the complete expectation of life. [1]
- ii) Expected present value is denoted by $\bar{a}_x$.

$$\bar{a}_x = E [\bar{a}_{T_x}]$$

$$\bar{a}_x = \int_0^{\infty} \bar{a}_{t|} {}_t p_x \mu_{x+t} dt \quad \text{Equation 1}$$

$$\text{Now: } \bar{a}_{t|} = \int_0^t e^{-\delta s} ds$$

$$\text{Hence: } \frac{d}{dt} \bar{a}_{t|} = e^{-\delta t} = v^t$$

Using Integration by part (pg 3 of tables)

$$\int_0^{\infty} u \frac{dw}{dt} dt = [uw]_0^{\infty} - \int_0^{\infty} w \frac{du}{dt} dt$$

If we set:

$$u = \bar{a}_{t|} \quad \text{and} \quad \frac{dw}{dt} = {}_t p_x \mu_{x+t}$$

so that:

$$\frac{du}{dt} = v^t \quad \text{and} \quad w = -{}_t p_x$$

Hence we have Equation 1 now:

$$\bar{a}_x = \int_0^{\infty} \bar{a}_{t|} {}_t p_x \mu_{x+t} dt$$

$$\bar{a}_x = -[\bar{a}_{t|} {}_t p_x]_0^{\infty} + \int_0^{\infty} v^t {}_t p_x dt$$

Hence

$$\bar{a}_x = \int_0^{\infty} v^t {}_t p_x dt \quad [4]$$

[5 Marks]

Solution 6:

i)

$$P = 50,00,000 * (A_{45} - v^{20} * {}_{20}p_{45} * A_{65})$$

$$P = 50,00,000 * (0.27605 - 1.04^{-20} * 8821.2612 * 0.52786 / 9801.3123)$$

$$P = \text{INR } 2,96,152$$

For 10,000 policies

$$= 10,000 * P$$

$$= \text{INR } 2,96,15,20,000 \quad [2]$$

ii) Per policy reserves at the end of 1st year:

$$\begin{aligned} V_1 &= 50,00,000 * (A_{44} - v^{19} * {}_{19}p_{44} * A_{63}) @ 6\% \\ &= 50,00,000 * (0.15146 - (1.06^{-19}) * (9037.3973/9814.3359) * 0.37091) \\ &= 50,00,000 * 0.0385741 \\ &= \text{INR } 1,92,870 \end{aligned}$$

$$\text{Mortality Profit} = \text{EDS} - \text{ADS}$$

$$\text{Here DSAR} = \text{SA} - V_1 = 50,00,000 - 1,92,870 = 48,07,130$$

$$\begin{aligned} \text{EDS} &= 10,000 * q_{45} * 48,07,130 \\ &= 10,000 * 0.001465 * 48,07,130 \\ &= 7,04,24,400 \end{aligned}$$

$$\text{ADS} = 10 * 48,07,130$$

$$\text{Mortality Profit} = 2,23,53,150 \text{ INR} \quad [5]$$

[7 Marks]

Solution 7:

i) Under with profit policies the contract between policyholder and the company is such that, whatever is the surplus arising on the underlying fund, would be shared between policyholders and shareholders in a predetermined manner. Now, as the surplus arises over future years the same is shared with policyholders in form of bonuses and hence the need of bonuses under with profit policies. [1]

ii) Investment Surplus
Expense Surplus
Mortality Surplus
Lapse / Surrender Surplus
Reinsurance Surplus [2]

iii) Terminal bonus could be a strategy to build lower guarantees over the lifetime of with profit policies. For example if the proportion of RB and TB is skewed towards TB then it would mean lower guarantees to Policyholders and hence the companies would have greater freedom in terms of choosing the investment assets. By doing this they can take risk of investing in real assets like equities and real estate which can lead to better final returns to policyholders, which would be distributed through higher TB. [2]

[5 Marks]

Solution 8:

i) Let Monthly Premium be P

$$\text{EPV of Premiums} = 12 * P * \ddot{a}_{50:107}^{(12)} \text{ at } 6\%$$

$$\text{Now: } \ddot{a}_{50:107}^{(12)} = \ddot{a}_{50:107} - (11/24) * (1 - {}_{10}p_{50} * v^{10})$$

$$\text{Now } \ddot{a}_{50:107} = 7.694 \text{ solving it we get}$$

$$\begin{aligned} \text{So: } \ddot{a}_{50:107}^{(12)} &= 7.694 - (11/24) * (1 - 1.06^{10} * (9287.2164/9712.0728)) \\ &= 7.694 - (11/24) * (1 - 0.533968) \\ &= 7.4804 \end{aligned}$$

Hence:

$$\begin{aligned} \text{EPV of Premiums} &= 12 * P * 7.4804 \\ &= 89.7648 * P \end{aligned}$$

$$\begin{aligned} \text{EPV of benefits} &= 10,00,000 * (q_{50} * v + 1.019231 * v^2 * {}_1|q_{50} + \dots \\ &\quad + 1.019231^{14} * v^{15} * {}_{14}|q_{50} \\ &\quad + 1.019231^{15} * v^{15} * {}_{15}p_{50}) \\ &= 10,00,000 * (1/1.019231) * (q_{50} * v * 1.019231 + 1.019231^2 * v^2 * {}_1|q_{50} + \\ &\quad \dots + 1.019231^{15} * v^{15} * {}_{14}|q_{50} + 1.019231^{15} * v^{15} * {}_{15}p_{50}) \\ &\quad + 10,00,000 * (1/1.019231) * v^{15} * {}_{15}p_{50} * (1.019231^{16} - 1.019231^{15}) \\ &\quad (\text{All at } 6\%) \end{aligned}$$

Converting to 4% (combination of 1/019231 and 6 % leads to 4%)

$$\begin{aligned} \text{EPV of benefits} &= 10,00,000 * (1/1.019231) * (A_{50:157} \text{ at } 4\%) \\ &\quad + 10,00,000 * v^{15} (\text{at } 6\%) * {}_{15}p_{50} * (1.019231^{15} - 1.019231^{14}) \end{aligned}$$

$$\text{Now using } A_{50:157} = 0.56719 \text{ we get}$$

$$\text{EPV of benefits} = 5,66,004.1$$

(Above step involves converting from 6% series to 4% by taking a common factor of 1.019231 outside the bracket).

EPV of Expense and Commission =

$$\begin{aligned} &(20 + 35)/100 * 12 * P + (2 + 1.5)/100 * {}_1p_{50} * 1.06^{-1} * 12P * \ddot{a}_{51:97}^{(12)} \\ &+ 2,500 + 400 * {}_1p_{50} * 1.06^{-1} * \ddot{a}_{51:147} \text{ at } 6\% \end{aligned}$$

$$= 9.3333 P + 6,115.06$$

Now equating EPV of Premiums = EPV of benefits + EPV of Expense and commission

$$89.7648 * P = 5,66,004.1 + 9.3333 * P + 6,115.06$$

$$80.4315 * P = 5,72,119.16$$

We get $P = 7,113.12$ INR

[6]

ii) Sum Assured at the end of 10th year + Accrued bonuses

$$X = 10,00,000 * 1.02^{10}$$

$$X = 12,18,994$$

PV of future benefits at the end of 10th Year =

$$= X/1.05 * q_{60} + X/1.05^2 * 1.04 * {}_1|q_{60} + X/1.05^3 * 1.04^2 * {}_2|q_{60}$$

$$+ X/1.05^4 * 1.04^3 * {}_3|q_{60} + X/1.05^5 * 1.04^4 * {}_4|q_{60}$$

$$+ X/1.05^5 * 1.04^5 * {}_5p_{60}$$

$$= 0.952948 * X$$

$$= 11,60,774$$

PV of expenses = $400 * \ddot{a}_{60:57}$ at 5%

$$= 400 * (1 + 1.05^{-1} * l_{61}/l_{60} + 1.05^{-2} * l_{62}/l_{60} + 1.05^{-3} * l_{63}/l_{60} + 1.05^{-4} * l_{64}/l_{60})$$

$$= 1,787.39$$

Reserves required = PV of Benefits + PV of Expenses

$$= 11,62,561 \text{ INR}$$

[4]

[10 Marks]

Solution 9:

This can be divided into sub benefits:

Working in '000 (thousands)

We can break the benefits into three components:

A. $A = \text{Guaranteed annuity for first 5 years} = 12 * 50 * a_{57}^{(12)} \text{ at } 4\%$

B. $B = \text{Benefit if both alive at end of 5th year} =$

$${}_5p_{65}(M) * {}_5p_{64}(F) * v^5 * [50 * 12 * a_{70(m):69(f)}^{(12)} + 25 * 12 * a_{70(m):69(f)}^{(12)}]$$

C. C = Benefit if only one is alive at the end of 5th year =

$${}_5p_{65}(M) * {}_5q_{64}(F) * v^5 * 2,500 + {}_5q_{65}(M) * {}_5p_{64}(F) * v^5 * 50 * 12 * a_{69(f)}^{(12)}$$

Now:

$$\begin{aligned} a_{57}^{(12)} @ 4\% &= i / i^{(12)} * a_{57} @ 4\% \\ &= [0.04 / \{(1.04^{(1/12)} - 1) * 12\}] * 4.4518 \\ &= (0.04 / 0.039285) * 4.4518 \\ &= 4.53283 \end{aligned}$$

$$\begin{aligned} a_{69(f)}^{(12)} &= (\ddot{a}_{69(f)} - 13/24) \\ &= (13.33 - 13/24) \\ &= 12.78833 \end{aligned}$$

$$\begin{aligned} a_{70(m):69(f)}^{(12)} &= [(\ddot{a}_{70(m)} - 13/24) + (\ddot{a}_{69(f)} - 13/24) - (\ddot{a}_{70(m):69(f)} - 13/24)] \\ &= (11.562 + 13.33 - 9.932 - (13/24)) \\ &= 14.41833 \\ &\text{(Where } \ddot{a}_{70(m):69(f)} = 9.932) \end{aligned}$$

$$\begin{aligned} a_{70(m):69(f)}^{(12)} &= (\ddot{a}_{70(m):69(f)} - 13/24) \\ &= (9.932 - 13/24) \\ &= 9.390333 \end{aligned}$$

Putting all these in above benefits and looking up other probabilities from given mortality tables we get:

$$A = 600 * 4.53283 = 2719.698$$

$$\begin{aligned} B &= (9238.134 / 9647.797) * (9473.745 / 9742.64) * \\ &\quad (50 * 12 * 14.41833 + 25 * 12 * 9.390333) / 1.04^5 \\ &= 0.95753 * 0.9724 * (600 * 14.41833 + 300 * 9.390333) / 1.04^5 \\ &= 8776.514 \end{aligned}$$

$$\begin{aligned} C &= 0.95753 * (1 - 0.9724) * 2500 / 1.04^5 \\ &\quad + (1 - 0.95753) * 0.9724 * 50 * 12 * 12.788 / 1.04^5 \\ &= 314.74839 \end{aligned}$$

$$PV \text{ of total benefits} = (A + B + C) * 1000 = \text{INR } 1, 18, 10,960$$

[10 Marks]

Solution 10:

i)

a) CMR is total number of deaths in the area divided by total population.

b) DSMR is the CMR that would have been applied in the area had the population structure been the same as that of the standard population .

c) ISMR is CMR multiplied by Area Comparability Factor (ACF). ACF is CMR for the standard population divided by the CMR for the standard population were it to have the same population structure as the area.

[4]

ii)

CDR of the country is as below:

$$\text{CDR}^* = 61503/11800000 = 0.005212$$

CDR of the country assuming population of country is like Area 10 is as below:

$$\begin{aligned}\text{CDR}^\# &= (522/3200000 \cdot 800 + 2522/4000000 \cdot 1200 + 20234/4200000 \cdot 1000 + 38225/400000 \cdot 300) / 3300 \\ &= 34.37/3300 \\ &= 0.01041\end{aligned}$$

$$\text{ACF} = \text{CDR}^* / \text{CDR}^\# = 0.5003$$

$$\text{ISMR} = \text{ACF} \cdot \text{CDR} = 0.5003 \cdot 32356/3300000 = 0.004906$$

[6]

[10 Marks]**Solution 11:**

The expected present value of the benefit payable on death from the healthy state is:

$$50000 \int_0^{20} {}_t p_{40}^{HH} \mu_{40+t} e^{-\delta t} dt$$

The expected present value of the benefit payable on partial disability is :

$$50000 \int_0^{20} {}_t p_{40}^{HH} \sigma_{40+t} e^{-\delta t} dt$$

Since its not possible to return to healthy state, ${}_t p_{40}^{HH}$ is evaluated as below

$${}_t p_{40}^{HH} = \frac{{}_t p_{40}^{HH}}{{}_t p_{40}^{HH}} = e^{-(\mu+\sigma)t} = e^{-0.008t}$$

Substituting in the values given in the question using $\delta = \log 1.05 = 0.048790$, expected present value of the benefit to be

$$\begin{aligned}
 & 50000 \int_0^{\infty} {}_t p_{40}^{HH} \mu_{40+t} e^{-\delta t} dt + 50000 \int_0^{\infty} {}_t p_{40}^{HH} \sigma_{40+t} e^{-\delta t} dt \\
 & 50000 \int_0^{\infty} e^{-0.008t} 0.005 e^{-0.048790t} dt + 50000 \int_0^{\infty} e^{-0.008t} 0.003 e^{-0.048790t} dt \\
 & 400 \int_0^{\infty} e^{-0.056790t} dt \\
 & \frac{400 e^{-0.056790t}}{-0.05679} \Bigg|_0^{\infty} \\
 & = 4781.3629
 \end{aligned}$$

[10 Marks]

Solution 12:

i)

The completed table is as below:

Year	Premium	Expenses	Interest	Expected cost of claims	Expected cost of increasing reserves (Net of interest earned on reserves)	Profit vector
1	1,530.00	(50.00)	88.80	(50.00)	(1,569.80)	(51.00)
2	1,530.00	(13.30)	91.00	(50.00)	(1,536.70)	21.00
3	1,530.00	(20.57)	90.57	(5,000.00)	3,445.00	45.00

- Interest in 1st year = $(1,530 - 50) \times 0.06 = 88.8$
- Expected cost of claims in first year: $5,000 \times 0.01 = 50$
- Expected cost of claims in second year: $5,000 \times 0.01 = 50$
- Expected cost of claim in 3rd year = 5,000 since all policies are in force at start of the year. These are negative entries in the table.
- Increase in reserve at the end of 1st year is the balancing figure = 1569.80
Reserve at the end of 1st year = $1,569.80 / 0.99 = 1,585.66$
Reserves at the end of 2nd year = $195 / 0.06 = 3,250$

6. Increase in reserve at the end of 2nd year = $3250 \times 0.99 - 1585.66 \times 1.06 = 1536.7$

7. Expenses in 2nd year = $1530 - (21 + 50 + 1536.7) / 1.06 = 13.3$

8. Release of reserve at the end of 3rd year is: $-_2V \times 1.06 = -3,250 \times 1.06 = -3,445$

9. Expenses in 3rd year = $1530 - (45 + 5,000 - 3,445) / 1.06 = 20.57$

10. Interest earned in 2nd year = $(1,530 - 13.3) \times 0.06 = 91$

11. Interest earned in 3rd year = $(1,530 - 20.57) \times 0.06 = 90.57$ [10]

ii)

IRR is the interest rate that satisfies the equation

$$0 = -51 \cdot v + 21 \cdot 0.99 \cdot v^2 + 45 \cdot 0.99^2 \cdot v^3$$

$$0 = -51v + 20.79v^2 + 44.10v^3$$

Dividing both sides by v,

$$0 = -51 + 20.79v + 44.10v^2$$

$$x = -b \pm \sqrt{b^2 - 4ac} / 2a$$

$$\begin{array}{ll} V & = 0.8652 \\ \text{so } i & = 15.58\% \end{array}$$

Note: Equation can be solved by trial and error as well as by interpolation. [2]

iii)

IRR will increase

This is because the profits in the later years will be released soon, which will increase their value

[2]

iv)

$$NPV = -51v + 21 \cdot 0.99 v^2 + 45 \cdot 0.99^2 v^3$$

$$= -51v + 20.79 v^2 + 44.1045 v^3$$

$$= -51/1.07 + 20.79/ (1.07)^2 + 44.1045 / (1.07)^3$$

$$= -47.66355 + 18.15879 + 36.00241$$

$$= 6.49765 \text{ or } 6.50 \text{ (approx.)} \quad [2]$$

v)

NPV will decrease

Profits will be released later which will decrease the present value

[2]

[18 Marks]

Solution 13:

i)

Unit fund represents the value of unit of the policyholder at any time.

It does not necessarily represent the amount that the policyholder is entitled to at any point. For example, on surrender, policyholder will get the surrender value which is the proportion of unit fund.

On death, maturity or surrender, the units held will be used to pay the benefit.

Non-unit fund will arise from the following sources:

- a. premium less cost of allocation i.e. the difference between the premium paid and the amount invested in the unit fund
- b. expenses incurred by the insurer
- c. interest earned on non-unit fund
- d. FMC taken from unit fund
- e. Cost of outgoes if outgo amount is greater than the value of units
- f. Profit on surrender

[6]

ii)

The life company sets up non unit reserves if the overall cash flow in any month other than month one is negative without reserves. The policy should be self-financing after month one.

Reserves are set up early so that money can be released as and when required to eliminate the negative cash flow. Setting up reserves will result in delay of profits but gives the comfort to the company that capital is sufficient to support the policy throughout the term.

[3]

[9 Marks]
